

Food Insecurity and Youth Depression Risk in the United States

Author

Ao Xu

Project repository

<https://github.com/bobaoxu2001/Food-insecurity>

Background

Food insecurity is a core social determinant of health. Adolescents exposed to household food insecurity may experience chronic stress, poorer diet quality, and psychosocial strain that contribute to depression risk.

Research question

Does household food insecurity increase the risk of depressive symptoms among adolescents and youth in the U.S.?

Data source

National Health and Nutrition Examination Survey (NHANES), pooled cycles 2005-2006 to 2021-2022 (8 cycles), using DEMO, DPQ (PHQ-9), FSQ, and BMX modules.

Literature Review and Methods

Literature review

Prior epidemiologic studies consistently report that food insecurity is associated with worse mental health outcomes in both adults and youth. Mechanisms include material deprivation, family stress, and reduced ability to maintain stable healthy routines.

Study design

Cross-sectional observational analysis using nationally representative NHANES survey data. Primary exposure was food insecurity status (low/very low vs full/marginal), harmonized from FSQ summary variables (FSDHH and, when needed, FSDAD). Primary outcome was moderate-to-severe depressive symptoms (PHQ-9 ≥ 10).

Population

Participants aged 12-19 years. Initial youth sample size: 11,659; complete-case analytic sample: 2,081.

Covariates

Age (years), gender (male vs female), family income-to-poverty ratio (INDFMPIR), and BMI (kg/m^2).

Model

Weighted logistic regression: Depression $\sim$ Food_Insecurity + Age + Gender + Income + BMI. MEC weights were rescaled by the number of pooled cycles.

Data Processing and Statistical Analysis

Data processing

Raw NHANES XPT files were downloaded directly from CDC public endpoints and merged by SEQN. PHQ items were cleaned to valid scores (0-3), and records with special missing codes were excluded from complete-case modeling. Food security status used FSDHH when available, with FSDAD harmonized for cycles where FSDHH was not released.

Missing-value handling

Complete-case analysis was used for outcome, exposure, and covariates in the main model. This approach preserves model interpretability but may introduce selection bias if missingness is not random.

Main finding (adjusted association)

Food insecurity was associated with higher odds of depressive symptoms: OR=3.04, 95% CI [2.13, 4.34], $p < 0.0001$.

Prevalence contrast

Weighted depression prevalence was 7.0% in food-secure youth versus 17.3% in food-insecure youth.

Results Figure 1

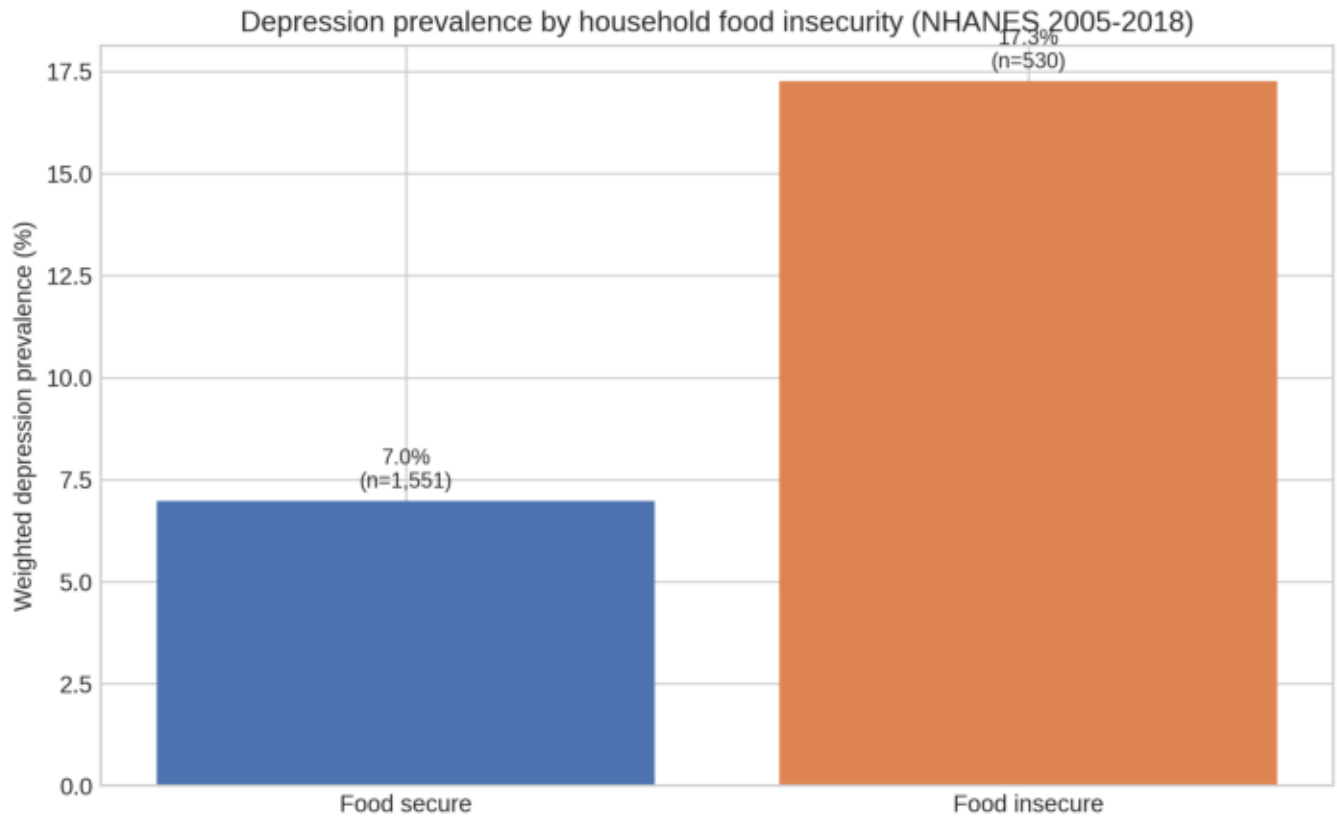


Figure 1. Weighted prevalence of depressive symptoms by household food insecurity status.

Results Figure 2

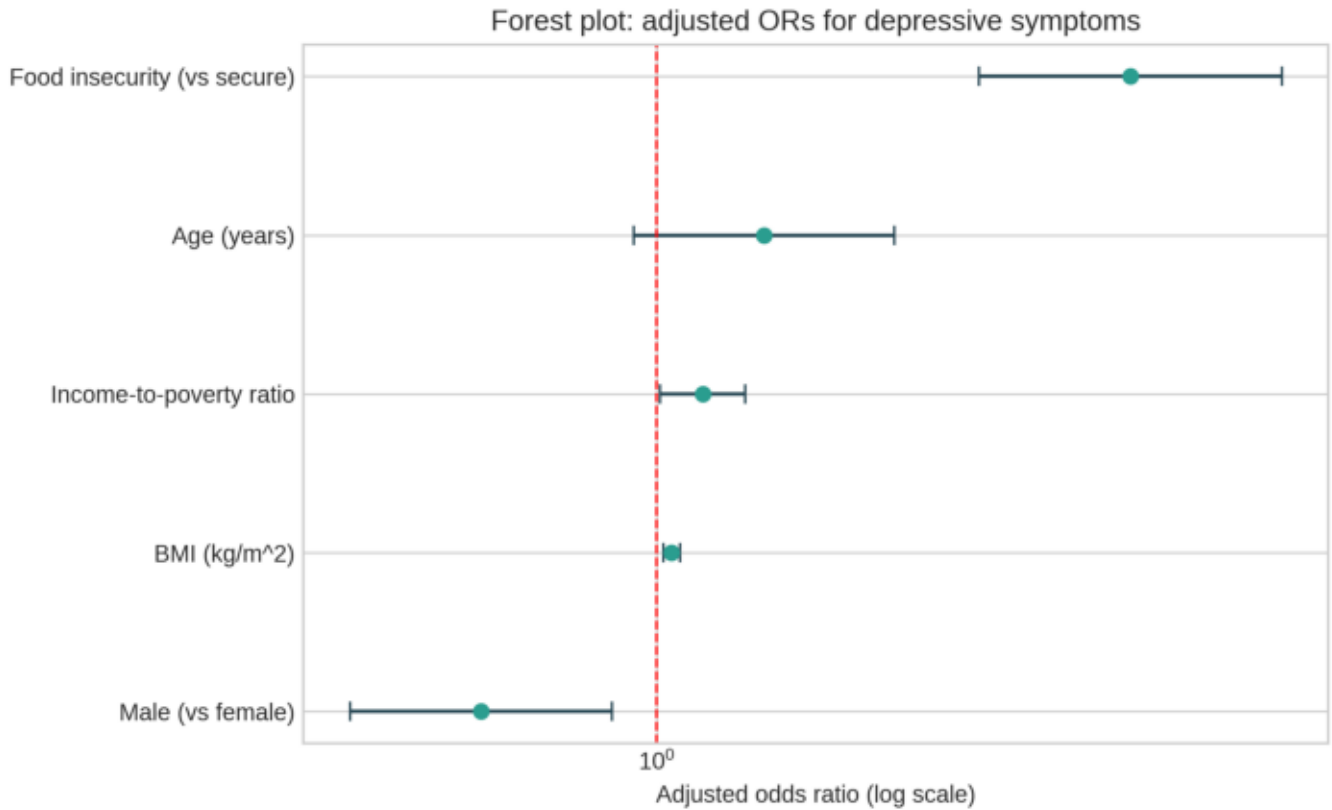


Figure 2. Forest plot of adjusted odds ratios from the multivariable weighted logistic regression.

Results Figure 3

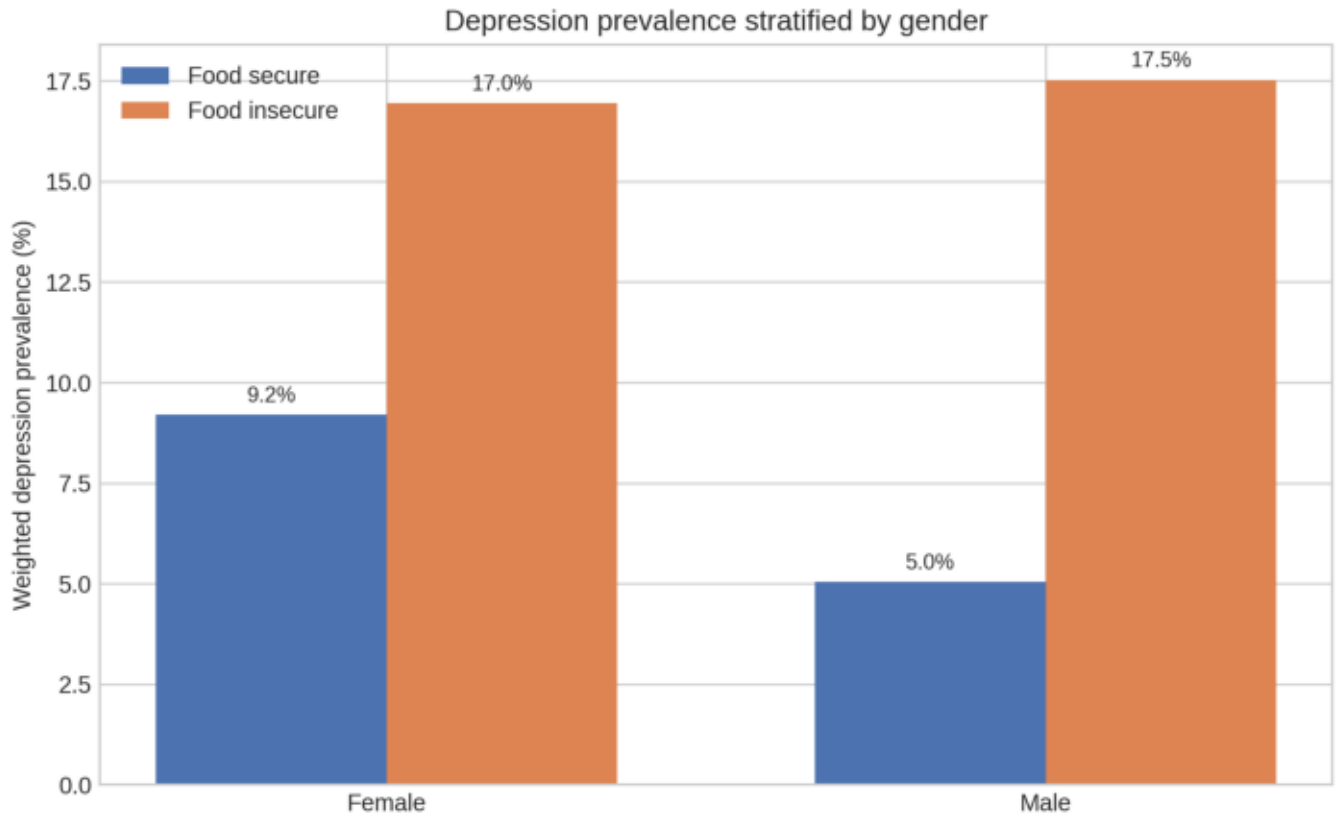


Figure 3. Weighted depression prevalence by food security status, stratified by gender.

Results Figure 4

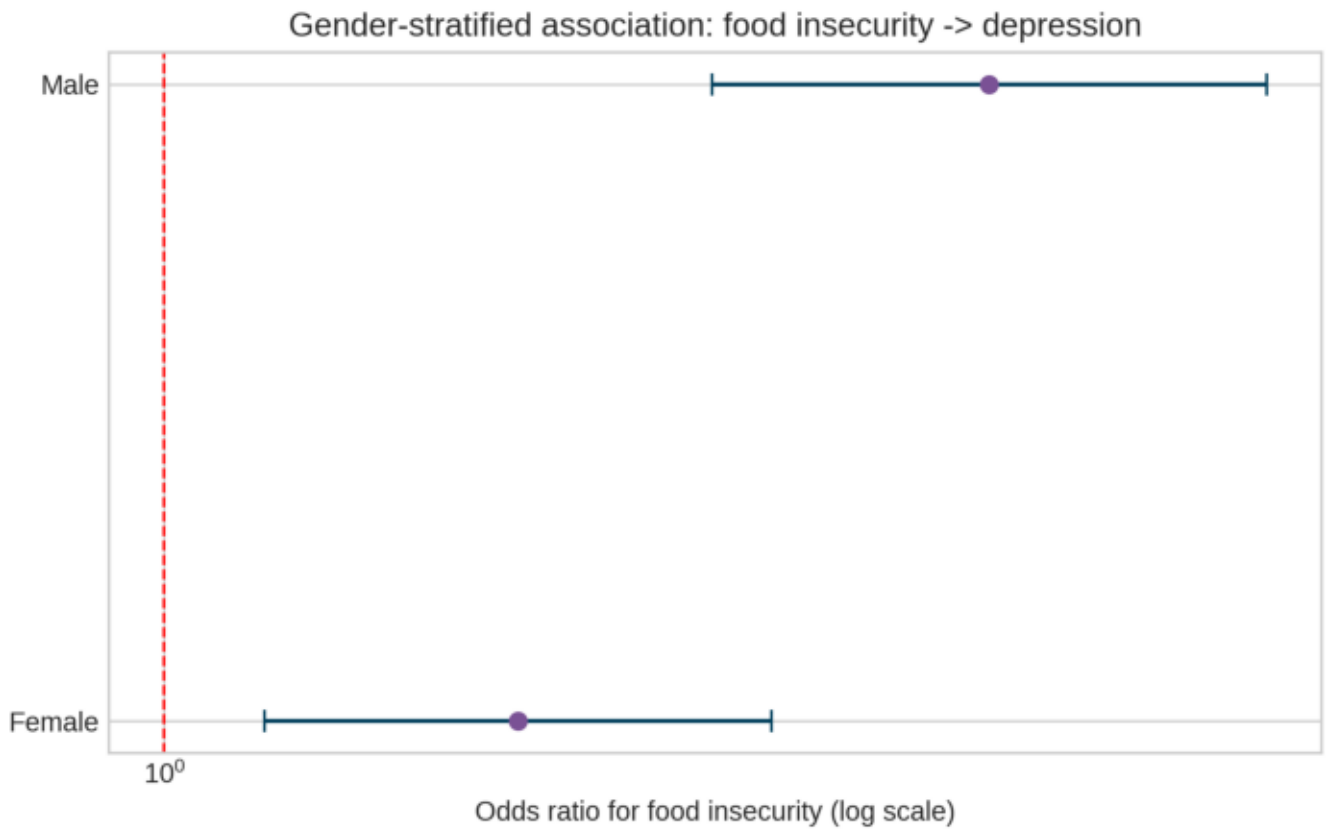


Figure 4. Gender-stratified adjusted odds ratios for the association between food insecurity and depression.

Key Regression Estimates

Adjusted odds ratios from the weighted multivariable logistic model.

Predictor	OR	CI_Lower	CI_Upper	p_value
Age (years)	1.2863	0.9471	1.7469	0.1070
BMI (kg/m^2)	1.0359	1.0162	1.0560	0.0003
Food insecurity (vs secure)	3.0414	2.1314	4.3400	0.0000
Income-to-poverty ratio	1.1140	1.0084	1.2305	0.0335
Male (vs female)	0.6618	0.4866	0.9001	0.0085

Gender-Stratified Model

Association between food insecurity and depression within each gender stratum.

Gender	N	OR_FoodInsecure	CI_Lower	CI_Upper	p_value
Female	1007	1.9734	1.2128	3.2110	0.0062
Male	1074	4.8747	2.8628	8.3003	0.0000

Discussion, Implications, and Future Research

Discussion

The analysis indicates a robust positive association between household food insecurity and depressive symptoms among U.S. youth, even after adjustment for demographics and socioeconomic indicators.

Public health implications

Policies that reduce household food insecurity (SNAP optimization, school meal access, and targeted community food support) may provide mental health benefits in addition to nutritional gains.

Limitations

Cross-sectional NHANES design cannot establish causality. Complete-case analysis may introduce bias. Residual confounding and measurement error remain possible.

Future research

Future work should test longitudinal pathways, mediation through diet quality and family stress, and effect modification by race/ethnicity, geography, and social support.

Reproducibility

All code, scripts, and outputs are reproducible from the project repository:
<https://github.com/bobaoxu2001/Food-insecurity>